

7 (a)	Fig. 6.1(a): approximately circular wavefronts	M1	
	centred on gap	A1	
	constant wavelength (allow this in (a) or (b))	B1	
	Fig. 6.1(b): wavefronts plane at centre	M1	
	curved at edges	A1	[5]
(b)	$\theta = \frac{1}{2}(162 - 136) = 13^\circ$	C1	
	$d \sin \theta = n\lambda$	C1	
	$d \sin 13 = 2 \times 630 \times 10^{-9}$	C1	
	$d = 5.6 \times 10^{-6} \text{ m}$	A1	[4]
	(Use of $\theta = 162^\circ$ or 136° , max 2/4)		
(c)	e.g. more slits for light to pass through		
	narrow so more diffracted light and 'off-axis' fringes clearer	B1	[1]